Supporting Information

Self-organizing dominance hierarchies generate the diversity of rank–stress profiles seen in social animals

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Appendix S1. Full model description (ODD protocol)

This appendix describes the model as implemented, following the ODD protocol (Grimm et al. 2020). Unless stated otherwise, every quantity is dimensionless. Defaults are the values in the simulation code. The settings at the three base points used in the main text (A, B, C) and the values explored in each experiment are listed in Table S1. Notation: i is the focal individual (actor), j a partner (recipient) and k an asset; $\sigma(x) = 1/(1+e^{-x})$; $(x)_+ = \max(x, 0)$.

1. Purpose and patterns

The model asks whether a near-transitive dominance hierarchy can emerge from local, experience-based decisions among initially identical individuals. Where one emerges, it asks which profiles of physiological burden across rank the same decisions generate. The model is built so that both the hierarchy and the burden are outputs. Individuals start exchangeable. Nobody is assigned a rank, and no behavioural or physiological parameter depends on status. Rank is computed only by the analyst, from the event log, after the run. It never enters any transition, belief, utility, policy or physiology equation. Load is never transmitted along network edges. An individual's load changes only through the events it experiences and its own energy state.

The model is evaluated against three groups of patterns.

1. **Structure.** The patterns are linear, near-linear, nontransitive, egalitarian, unstable and fragmented dominance structure, judged by triad transitivity, Landau's h', observation coverage, persistence and held-out prediction.
2. **Rank–load profiles.** The patterns reported in field studies are graded subordinate-high, dominant-high, U-shaped, inverted-U and flat profiles, with or without an exceptional alpha.
3. **Channels.** The top-versus-bottom load gap is decomposed into its input channels (defeat, intimidation, effort, deficit, unpredictability).

2. Entities, state variables and scales

Individuals. There are N individuals ($N = 10$ by default; 6–20 in the experiments). Each slot i carries the state variables below. Slot identifiers carry no strategic content: no agent uses them to choose.

Variable	Domain	Meaning
E_i	$[0, 1]$	energy reserve
z_i	≥ 0	fast excess activation
A_i	≥ 0	slow allostatic load (recoverable burden)
B_i	≥ 0	received-support trace
R_i	$[0, 1]$	functional reserve (only under coupling mechanism F4)
mode	—	forage, rest, withdrawn, holding a food patch, consorting
busy	—	none, fighting, grooming, cooperating (with partner and end time)
r_i	$\{0, 1\}$	mating readiness (1 for all individuals in season)
ledgers	≥ 0	cumulative intake, maintenance, effort, injury, waste and deficit D_i
beliefs	—	decaying evidence counts about each partner and pooled over partners (§7.4)
avoidance timers	—	time until which i avoids j
critic weights	$\mathbb{R}^7$	only under policy P4
subgroup	$\{1, \dots, n_{\text{sub}}\}$	only under fission–fusion

The model also records counters for injuries, accepted matings and skipped opportunities, and the per-channel cumulative physiological inputs.

Assets. There are K_F food patches, each held by at most one individual. Optional mating seasons add K_M consortship assets, which exist only in season. Dispersed food is available everywhere and cannot be monopolized. Patch relocation is not implemented.

Environment and group. The model has no explicit space. Contact is a weighted sampling of the other group members at each decision opportunity (§7.2). Avoidance reduces the weights. Under fission–fusion, sampling is restricted to the individual's current subgroup.

Scales. The time unit is the mean interval between one individual's decision opportunities. Each individual's opportunities follow a Poisson process of rate $\nu = 1$, so total activity scales with N, not N^2. Runs last

$T_{\text{run}} = 3000$ time units (9000 at $N = 20$). Node states are recorded on a grid of spacing $\Delta t_{\text{rec}} = 1$. Physiological time scales are set relative to this unit: $1/\lambda_z = 1$ for activation and $1/\delta_{A0} = 100$ for load, so load integrates activation over about 100 decision intervals. Energy, activation, load and utility are dimensionless. Utility is measured in energy equivalents. Neither z nor A is a hormone concentration.

3. Process overview and scheduling

The model is a continuous-time discrete-event simulation. The event queue holds the following event types:

- decision opportunities, a superposed Poisson process of rate $N\nu$ with the actor drawn uniformly;
- the scheduled ends of fights, grooming and cooperation;
- mating attempts by idle consortship holders;
- season boundaries;
- random membership turnover and the optional targeted removal;
- subgroup moves under fission–fusion;
- recording-grid times.

Continuous state is advanced exactly and lazily, one individual at a time, only when that individual is touched by an event. The advanced state comprises energy, belief decay, the exposure times used by hazard beliefs, physiology and the reserve.

Decision opportunity. The random stream used at each step is given in brackets.

1. Advance the actor to the current time. If it is busy, record a skipped opportunity and stop.
2. Draw the encounter set and the visible assets [contact], and build the feasible option set (§7.2).
3. Under P4 only, perform the actor's temporal-difference update for the interval since its previous decision.
4. Compute the value $Q_i(d)$ of every feasible option. Under P4, add the learned continuation. Under P3D, add the deterrence value. Sample an option [choice].
5. If the option is social, advance the recipient, compute its response values and sample a response [response].
6. Resolve the interaction [contest]: the escape draw, or the fight outcome and injury.
7. Apply the energy ledger (effort, escape and injury costs), asset transfers, mode changes and busy flags, and schedule the end of the interaction.
8. Write exposure rows, apply physiological and support jumps, and update continuous inputs.
9. Update the beliefs of the participants, and of bystanders if enabled. Set avoidance timers.
10. Write the event row to the log.

All effects of an interaction are applied at the initiating event. The participants are then busy for the interaction's duration (d_F, d_G or d_C), and the scheduled end event only clears the busy flags and restores modes. An individual that still holds an asset resumes holding; every other participant enters forage mode.

Other events. Each first advances the affected individuals. A mating attempt draws acceptance [mating]. Season boundaries create or dissolve consortships. Turnover and removal replace an individual [shocks], and a subgroup move reassigns one [fission]. A grid time advances everyone and writes node rows.

4. Design concepts

Basic principles. The game combines four ideas: a finite energy budget with a solitary outside option (foraging alone); bounded-rational choice by expected subjective utility; Bayesian-style learning from one's own experience; and a linear activation–load model of allostasis driven by an auditable ledger of realized events. The model keeps four levels separate: (a) objective ecological transitions and fitness proxies; (b) agents' beliefs; (c) subjective utility; and (d) decision values and policies. The true win probability P_{ij}, the agent's forecast $\hat{p}_{ij}$ and the analyst's dominance estimate are distinct objects. Fitness proxies are recorded but never used for choice.

Emergence. The following are emergent: dominance structure; who holds patches and consortships; the network of directed outcomes, affiliation and avoidance; each individual's exposure profile; and therefore the rank–load profile and its channel composition. None is imposed.

Adaptation. At each opportunity an individual chooses among feasible actions, targets and efforts by softmax over expected utility (§7.6). A recipient chooses its response in the same way. Choices depend on the individual's energy (through the marginal value of energy), its holdings, the season and its beliefs.

Objectives. Each option is valued by the expected subjective utility of net energy, accepted matings, injuries, received safety events and received grooming, over a planning horizon H (§7.5). The baseline utility has no rank term and no load term.

Learning. Beliefs are decaying counts of the individual's own experience (rate γ), turned into Beta and Dirichlet posterior means (§7.4). Under P4, each individual also learns a linear critic by semi-Markov TD(0) (Sutton and Barto 2018). No individual learns from another's hidden state. Bystander learning, when enabled, uses only fights the bystander witnessed.

Prediction. Individuals forecast their win probability against each partner and the partner's effort, response and acceptance. They also forecast the hazard of losing a held asset, the rates at which they are attacked, and the probability that a received attack ends badly. Under P3D an individual also predicts how a partner's view of it, and hence the partner's challenge rate, would change after the interaction. All predictions use only the individual's own records.

Sensing. An individual observes its own energy, readiness, mode and holdings; the season calendar; which assets are held, by whom, and who is busy; its own encounters and the outcomes of its own interactions, including the partner's effort after a fight; and, as structural knowledge, ϕ and ϕ_W. It never observes another individual's energy, activation, load, support or reserve, global rank, or any meaning in identifiers. Efforts are committed simultaneously, so a recipient does not see the actor's effort when it responds.

Interaction. Interactions are direct and pairwise: challenges over held assets, aggression without a stake, grooming and cooperation. Positive interactions need the partner's acceptance. Attacks do not, but the recipient may submit, attempt escape or fight. Indirect interaction comes through competition for a fixed number of patches and consortships, and through avoidance, which lowers contact rates after a defeat or yield.

Stochasticity. Ten independent random streams are each used only for their named purpose: schedule, contact, choice, response, contest, mating, learning, shocks, observation and fission. Stream seeds are derived from the run seed and stream name by SHA-256, so draws on one stream never shift another's and results do not depend on parallel execution order. The learning stream is reserved (P4 updates are deterministic), and the observation stream serves only F1. Under *common random numbers*, replicate r of every point in a sweep uses the same seed.

Collectives. The group is the only collective, and the unit of replication. Fission–fusion subgroups (§7.9) restrict contact and do not make collective decisions.

Observation. All analysis is external to the simulation and follows frozen analysis plans; the main-text results use plan version 4. The event log records every proposal, response, contest, yield, affiliative event, exposure jump, mating attempt and membership change. Grid rows record each individual's state and cumulative channel inputs.

- **Windows.** After a burn-in of 1000, four windows of 500 are analysed. Each of the first three is classified, using the next window for persistence and held-out prediction. At $N = 20$ the burn-in is 3000 and the windows are 1500 each.
- **Structure classes.** Fights and yields are pooled as dominance evidence, and the first matching class is assigned:
  - *fragmented*: the contact graph is disconnected, or coverage < 0.6;
  - *Linear*: $h' \geq 0.8$, triangle transitivity above an orientation null ($p < 0.05$), coverage ≥ 0.8, and the held-out accuracy of a Bradley–Terry order at least that of dyad-specific majorities minus 0.02;
  - *nontransitive*: consistent dyads (DCI ≥ 0.7), with transitivity not above the null or 3-cycles above it;
  - *near-linear*: transitive, consistent and persistent (rank persistence ≥ 0.5), with $h' \geq 0.6$;
  - *egalitarian*: DCI < 0.5;
  - *unstable*: persistence < 0.5.
- **Run class.** The class shared by at least two of the three classified windows.
- **Ranks and profiles.** Ranks are David's scores. The rank–load profile rule is described in the main text.
- **Channel attribution.** For the linear physiology, window-mean load decomposes exactly by input channel.

5. Initialization

All individuals start identical:

- $E_i = E_0 = 0.5$, $z_i = 0$, $A_i = \bar{A} = 0$, $B_i = 0$ and $R_i = 1$;
- mode forage, holding nothing, not busy, with no avoidance;
- every evidence count at zero, so every forecast is at its prior (§7.4);
- under P4, critic weights at zero;
- all patches free.

If $K_M > 0$, the run starts at the beginning of a season. Under fission–fusion, initial subgroups are assigned uniformly [fission]. Under the controls, C1 status starts at 0 for all, and C3 draws its fixed dyadic win probabilities at $t = 0$ [contest]. Every run writes a manifest with its seed lineage, configuration and software revision.

6. Input data

None. The model uses no external or empirical data, and no time series drives it.

7. Submodels

7.1 Ecology and energy ledger

Food. A forager takes in dispersed food at rate g_s. A patch holder takes in $g_P = \rho_c g_s$, with $\rho_c \geq 1$. Scarcity is the ratio m/g_s of maintenance to dispersed intake: it is 0.8 at the defaults and 1.2 at base point A. With $m/g_s > 1$ a lone forager loses energy. For matched-supply comparisons, total supply $G_{\text{tot}} = (N - K_F)g_s + K_F\rho_c g_s$ is held fixed and ρ_c is set from K_F. This gives, for example, $G_{\text{tot}} = 0.475$ at $N = 10$, so that $K_F = 1$ gives $\rho_c = 10$.

Modes. Intake g_μ and expenditure m_μ depend on the mode.

Mode	Intake g_μ	Expenditure m_μ
forage (fallback, outside option)	g_s	m
rest	0	m_{rest}
withdrawn (refuge)	$\psi_W g_s$	m
holding a patch	$\rho_c g_s$	m
consorting	$\psi_M g_s$	$m + m_G$
busy: fight or grooming	0	m
busy: cooperation	$(1 + \sigma_c)g_s$, or $(1 + \sigma_c)\rho_c g_s/2$ if either partner holds a patch	m

Between events, $\dot{E}_i = g_\mu - m_\mu$. Bounds are handled without clipping. The time at which E_i reaches 0 is computed in closed form and the interval is split there. At $E_i = 0$ with $g_\mu < m_\mu$, unpaid maintenance accrues as a recorded deficit, $\dot{D}_i = m_\mu - g_\mu$. At $E_i = 1$, excess intake is recorded as waste. Instantaneous costs are charged at the event:

- effort $c_e e$ to each fighter;
- c_{esc} per escape attempt;
- E_{inj} per injury.

Any part of a cost exceeding E_i is added to D_i. After every event the simulator checks the conservation identity

$$E_i(t) = E_i(0) + \text{Intake}_i - \text{Basal}_i - \text{Effort}_i - \text{Injury}_i - \text{Waste}_i + D_i.$$

Time costs enter only as forgone intake.

Fitness proxies. These are recorded only and never used for choice. The survival proxy is $-\ln S_i = \mu_0 t + \mu_D T_i^{E=0} + \mu_I n_i^{\text{inj}}$, where $T_i^{E=0}$ is the time spent in deficit. The model also records the number of accepted matings M_i.

7.2 Actions, responses and outcomes

Contact. At an opportunity, the actor samples up to $n_{\text{enc}} = 2$ encountered individuals from those not busy [contact]:

1. Each withdrawn candidate is protected, and excluded, with probability ϕ_W.

2. Each remaining candidate has weight $1 - v$ if either individual avoids the other (an avoider holding a patch does not avoid), and weight 1 otherwise.
3. Candidates are drawn by weighted sampling without replacement (Efraimidis–Spirakis keys).

The actor also sees up to $n_{\text{vis}} = 2$ assets, drawn uniformly from the existing assets other than its own, together with their occupancy and holders.

Initiator actions. Effort is discrete, $e \in \{e_L, e_H\}$, and is chosen before the outcome.

- FORAGE, REST: always feasible.
- WITHDRAW: feasible when $\phi_W > 0$.
- KEEP: continue holding the current asset.
- OCCUPY(k): take a visible free asset.
- CHALLENGE(j, k, e): contest the holder j of a visible asset k, provided j is not busy.
- AGGRESS(j, e): attack an encountered individual without a stake.
- AFFILIATE(j): propose grooming.
- COOPERATE(j): propose joint foraging or patch sharing. Feasible only if neither partner is consorting and at most one holds a patch.

Holder rules:

- An individual holds at most one asset. OCCUPY and CHALLENGE are feasible only for non-holders.
- Choosing FORAGE, REST or WITHDRAW releases the held asset.
- A holder keeps its asset during AGGRESS, grooming and cooperation, and cannot be challenged while busy.
- Consortships can be occupied or challenged only in season.

Responses.

- To CHALLENGE or AGGRESS: SUBMIT, ESCAPE, or FIGHT with effort e_L or e_H.
- To AFFILIATE: ACCEPT, RECIPROCATE (grooming in both directions) or DECLINE.
- To COOPERATE: ACCEPT or DECLINE.

A declined proposal costs the proposer only the opportunity.

Outcomes.

- *SUBMIT.* To a CHALLENGE, the stake passes to the actor and the recipient enters forage mode. To an AGGRESS, the recipient keeps its mode.
- *ESCAPE.* The recipient pays c_{esc} and succeeds with probability ϕ [contest]. Escape has no duration. Under a CHALLENGE the stake is ceded whether or not escape succeeds. A failed escape counts as a yield that also received the attack.
- *FIGHT.* Both pay $c_e e$ and are busy for d_F. The winner keeps or takes the stake. The loser is injured with probability $\iota\, e_{\text{winner}}$, paying E_{inj}.
- *Accepted grooming.* Both are busy for d_G with no intake.
- *Accepted cooperation.* Both are busy for d_C at the cooperative intake rate.

Avoidance. This is a fixed rule, not a choice. After losing a fight to j, submitting to j or attempting escape from j, individual i avoids j for τ_V. Avoidance lowers the contact weight between the pair (above).

7.3 Contest success function

The objective probability that i beats j is a Tullock contest success function (Skaperdas 1996) with condition dependence:

$$P_{ij} = \frac{c_i e_i^\theta}{c_i e_i^\theta + c_j e_j^\theta} = \sigma\left(\theta \ln \frac{e_i}{e_j} + \zeta \ln \frac{\varepsilon_c + E_i}{\varepsilon_c + E_j}\right), \qquad c_i = (\varepsilon_c + E_i)^\zeta.$$

Here θ is decisiveness and ζ is condition dependence; $P_{ij} + P_{ji} = 1$. The outcome is drawn on the contest stream. The function provides two separately testable routes to order:

- **Belief route** ($\zeta = 0$). Outcomes are effort lotteries, and any structure must come from partner-specific beliefs and choices.
- **Resource-holding route** ($\zeta > 0$). Winning patches raises E, and higher E raises later win probability. Because E_i is a per-individual scalar, the objective win probabilities at equal efforts are strongly stochastically transitive (L15 holds for any positive strengths). It is earned through ecology, not assigned.

Agents never observe E_j, so the condition term never enters their forecasts.

Controls. These replace the primary contest function in comparison arms.

- **C1, scalar winner–loser comparator.** $P_{ij} = \sigma(\beta_s(s_i - s_j) + \theta \ln(e_i/e_j))$, with status updates $s_w \mathrel{+}= \Delta(1-P)$ and $s_l \mathrel{-}= \Delta(1-P)$, where P is the winner's pre-fight win probability. It is transitive by construction.
- **C3, fixed random dyadic.** $\operatorname{logit} P_{ij} = \operatorname{logit} U_{ij} + \theta \ln(e_i/e_j)$, with $U_{ij} \sim \mathrm{U}(0,1)$ drawn once and $U_{ji} = 1 - U_{ij}$.

7.4 Beliefs

Evidence counts. Each individual i holds counts about each partner j, built from its own experience:

- fight wins w_{ij} and losses l_{ij}, in either role;
- yields received s^+_{ij} (j submitted to or attempted escape from i) and yields given s^-_{ij};
- j's high-effort and low-effort uses against i;
- j's responses (submit, escape, fight) to i's challenges and, separately, to i's aggression;
- acceptances and declines of i's grooming and cooperation proposals, and reciprocation given acceptance.

Pooled counts over all partners give i's own baseline frequencies. They also record challenges received, assets lost and decayed holding time (separately for patches and consortships); aggression received and decayed time while not holding; received attacks and those that ended badly; and own mating attempts accepted and rejected. All counts decay as $e^{-\gamma \Delta t}$, applied lazily on read. Direct dominance evidence is $n_{ij} = w_{ij} + l_{ij} + s^+_{ij} + s^-_{ij}$. A dyad with $n_{ij} < n_U$ is *unobserved*, which is not the same as an observed equal relationship. Pooled frequencies use uniform priors, $(1+x)/(K+\sum)$ for K categories, so they equal $1/K$ with no experience.

Win forecast. A Beta posterior mean:

$$\hat{p}_{ij} = \frac{a_i + w_{ij} + \omega_S s^+_{ij} + w^{\mathrm{obs}}_{ij}}{2\alpha + w_{ij} + l_{ij} + \omega_S(s^+_{ij} + s^-_{ij}) + w^{\mathrm{obs}}_{ij} + l^{\mathrm{obs}}_{ij}}.$$

Here $a_i = \alpha$ by default. The effort-conditional forecast is $\hat{p}_{ij}(e, \hat{e}_j) = \sigma(\operatorname{logit} \hat{p}_{ij} + \beta_{\mathrm{opt}} + \theta \ln(e/\hat{e}_j))$, which assumes that agents know effort matters. The optimism bias β_{opt} is 0 by default.

Other forecasts.

- *Effort.* $\hat{\pi}^H_{ij} = (\alpha \bar{f}^H_i + f^H_{ij})/(\alpha + f^H_{ij} + f^L_{ij})$, where $\bar{f}^H_i$ is i's pooled high-effort frequency across opponents. Then $\hat{e}_j = e_L + \hat{\pi}^H_{ij}(e_H - e_L)$ is used as a point estimate.
- *Responses.* For each context (stake or no stake), the Dirichlet posterior mean $\hat{\pi}_{ij} = (\alpha_R \bar{R}_i + R_{ij})/(\alpha_R + \sum R_{ij})$, where $\bar{R}_i$ is i's pooled response frequency and $\alpha_R = 3$.
- *Acceptance and reciprocation.* $\hat{a}_{ij} = (2\bar{a}_i + Y_{ij})/(2 + Y_{ij} + N_{ij})$, where $\bar{a}_i$ is i's pooled acceptance frequency.
- *Mate acceptance.* $\hat{a}_i = (1 + Y^{\mathrm{mate}}_i)/(2 + Y^{\mathrm{mate}}_i + N^{\mathrm{mate}}_i)$.
- *Holding hazard*, by asset type τ: a Gamma–Beta estimate,

$$\hat{h}_i(\tau) = \frac{a_h + n^{\mathrm{ch}}_i(\tau)}{b_h + T^{\mathrm{hold}}_i(\tau)} \cdot \frac{\alpha + n^{\mathrm{lost}}_i(\tau)}{2\alpha + n^{\mathrm{ch}}_i(\tau)}.$$

  The first factor is the perceived attack rate while holding, $\hat{\lambda}_i(\tau)$.
- *Attack rate while free.* $\hat{\lambda}_i(\emptyset) = (a_h + n^{\mathrm{agg}}_i)/(b_h + T^{\mathrm{free}}_i)$, counting aggression received while foraging or resting.
- *Expected safety cost per received attack.* $\varsigma_i = s_S(\alpha + n^{\mathrm{bad}}_i)/(2\alpha + n^{\mathrm{att}}_i)$.

Exposure times accrue only while the individual is not busy.

Optional mechanisms coupling beliefs across dyads. These are off by default and are ablated separately.

- *Standing cue.* $\hat{s}_i = (1 + W_i)/(2 + W_i + L_i)$, where W_i is i's decayed number of fights won plus yields received over all partners, and L_i is fights lost plus yields given. With the cue on, the prior for every partner becomes $\mathrm{Beta}(2\alpha\hat{s}_i, 2\alpha(1-\hat{s}_i))$, that is, $a_i = 2\alpha\hat{s}_i$. The prior strength is unchanged.
- *Bystander learning.* Bystanders of a fight are the other members of the actor's encounter set. A bystander adds a pseudo-loss of weight ω_{obs} to its record against the winner and a pseudo-win against the loser. Pseudo-counts (w^{obs}, l^{obs}) enter $\hat{p}$ but not n_{ij}, so observation never makes a dyad directly observed.

Learning controls. C2 freezes every count at its prior. P2-pooled freezes the partner-specific counts but still learns the pooled ones.

7.5 Utilities and decision values

Utility. Energy is valued by a concave, state-dependent function, $\Phi(E) = E_h(1-e^{-E/E_h})$, so $\Phi'(E) = e^{-E/E_h}$: an energy change is worth more to a hungry individual. The utility of an outcome o is

$$u_i(o) = \Phi'(E_i)\,\Delta E_i(o) + s_M^u r_i\, n_i^{\text{mate}}(o) - s_H\, n_i^{\text{inj}}(o) - s_S\, n_i^{\text{saf}}(o) + s_G\, n_i^{\text{aff}}(o).$$

- ΔE is net energy. It already includes effort, forgone intake and injury energy, and at decision time it is the unclipped linear prediction.
- *Safety events* are received attacks that end badly for the recipient: a submission, a failed escape or a fight defeat.
- n^{aff} counts received grooming only. Received cooperation is valued through ΔE, to avoid double counting.
- The subjective safety cost s_S is an aversion valued at decision time. It does not read z or A.

Value components. All values are measured relative to remaining in forage mode, the outside option.

- *Horizon factor.* $\Lambda(h; H) = (1 - e^{-(\kappa+h)H})/(\kappa + h)$, with $\Lambda_0 = \Lambda(0; H)$.
- *Asset value.* $W_i(k) = \upsilon_i(k)\,\Lambda(\hat{h}_i(\tau_k); H_k)$. For a patch, $\upsilon = \Phi'(E_i)(\hat{\rho}_c - 1)g_s - \varsigma_i(\hat{\lambda}_i(F) - \hat{\lambda}_i(\emptyset))$, where $\hat{\rho}_c = \rho_c$ unless a misperceived richness is set (Table S1). For a consortship, $\upsilon = \Phi'(E_i)(\psi_M g_s - m_G - g_s) + s_M^u \mu_m \hat{a}_i - \varsigma_i(\hat{\lambda}_i(M) - \hat{\lambda}_i(\emptyset))$. The horizon is $H_k = H$ for patches and $\min(H, t^{\text{end}}_{\text{season}} - t)$ for consortships.
- *Forgone flow while busy.* $\ell_i = \Phi'(E_i) g_{\text{cur},i} + s_M^u \mu_m \hat{a}_i \mathbf{1}[i \text{ consorting}]$.
- *Expected fight cost.*

$$K_i(e; j) = \ell_i d_F + \Phi'(E_i) c_e e + (1 - \hat{p}_{ij}(e, \hat{e}_j))[s_S + \iota \hat{e}_j (s_H + \Phi'(E_i) E_{\text{inj}})].$$

- *Current holding term.* $\mathcal{H}_i = W_i(k_i)$ if i holds k_i, and 0 otherwise.

Initiator values.

Option	$Q_i(d)$
FORAGE	0
REST	$\Phi'(E_i)(m - m_{\text{rest}} - g_s)\Lambda_0$
WITHDRAW	$[\Phi'(E_i)(\psi_W - 1)g_s + \varsigma_i \phi_W \hat{\lambda}_i(\emptyset)]\Lambda_0$
KEEP, OCCUPY(k)	$W_i(k)$
CHALLENGE(j, k, e)	$(\hat{\pi}_{ij}^{\text{sub}} + \hat{\pi}_{ij}^{\text{esc}}) W_i(k) + \hat{\pi}_{ij}^{\text{fight}}[\hat{p}_{ij}(e, \hat{e}_j) W_i(k) - K_i(e; j)]$
AGGRESS(j, e)	$\mathcal{H}_i - \hat{\pi}_{ij}^{\text{fight}} K_i(e; j)$ (no-stake response forecast)
AFFILIATE(j)	$\mathcal{H}_i + \hat{a}_{ij}^G [s_G \hat{\rho}_{ij}^{\text{rec}} - \ell_i d_G]$
COOPERATE(j)	$\mathcal{H}_i + \hat{a}_{ij}^C d_C [\Phi'(E_i) g_{ij}^C - \ell_i]$

Here g_{ij}^C is the cooperative intake of §7.1.

Recipient values. These are relative to the recipient's status quo, with stake value $W = W_j(k)$ for a CHALLENGE and $W = 0$ for an AGGRESS:

$$Q_j(\text{SUBMIT}) = -W - s_S, \qquad Q_j(\text{ESCAPE}) = -W - \Phi'(E_j) c_{\text{esc}} - (1 - \phi) s_S,$$

$$Q_j(\text{FIGHT}(e')) = -(1 - \hat{p}_{ji}(e', \hat{e}_i)) W - K_j(e'; i).$$

For grooming, $Q_j(\text{ACCEPT}) = Q_j(\text{RECIPROCATE}) = s_G - \ell_j d_G$. For cooperation, $Q_j(\text{ACCEPT}) = d_C[\Phi'(E_j) g_{ji}^C - \ell_j]$. Declining is worth 0.

Consequently, ESCAPE beats SUBMIT exactly when $\phi s_S > \Phi'(E_j) c_{\text{esc}}$; AGGRESS has no positive immediate value, so myopic policies choose it only through noise; and at $s_G = 0$ grooming has negative immediate value.

7.6 Policies

P2 (primary; myopic softmax). The option is chosen with probability

$$\pi_i(d) = \frac{\exp(Q_i(d)/T)}{\sum_{d' \in \mathcal{D}_i} \exp(Q_i(d')/T)}, \qquad T = \tau_T U_{\text{ref}}, \quad U_{\text{ref}} = \Phi'(E_0) g_s H.$$

The choice is over the enumerated (action, target, effort) options, with $H = 10$. Responses use the same rule. Tying the temperature to the reference utility makes τ_T a relative noise level. Softmax over enumerated options weights action types by their number of targets and efforts. This weighting is declared, not corrected. Softmax convergence is not a Nash equilibrium.

P3 and P3-T (long horizon). P3 is P2 with $H = 100$. Because T scales with H, P3 also changes the absolute noise. P3-T evaluates values at $H = 100$ but keeps the temperature scale at $H = 10$ ($T_H = 10$).

P4 (learned continuation). The value of an option is

$$Q_i(d) = Q_i^{\text{imm}}(d) + e^{-\kappa/\nu} \sum_b \hat{\pi}_b \hat{V}_i(x'_b).$$

- Q^{imm} is the P2 value computed with the immediate horizon $H_{\text{imm}} = 1$.
- The branches b are weighted by i's own response and win forecasts.
- In the predicted post-outcome states x'_b only holdings and the standing $\hat{s}_i$ change (by one win or one loss). Energy is not projected.
- The critic is linear, $\hat{V}_i = w_i \cdot \varphi$, with features $\varphi = (1, E_i,$ holds a patch, holds a consortship, $\hat{s}_i,$ in season). Variant P4b adds i's mean win forecast over all others.
- *Learning.* At each of i's non-skipped opportunities the critic takes a semi-Markov TD(0) step (Sutton and Barto 2018). With s_n the state at the previous opportunity and s_{n+1} the current one, $\delta = R + e^{-\kappa \Delta t} \hat{V}(s_{n+1}) - \hat{V}(s_n)$ and $w \leftarrow w + \alpha_w \delta \varphi(s_n) / \|\varphi(s_n)\|^2$.
- *Reward.* R is the exact change $\Phi(E_{t_{n+1}}) - \Phi(E_{t_n})$ plus non-energy utility accrued in the interval: $-s_S$ per safety event, $-s_H$ per injury, $+s_M^u$ per accepted mating and $+s_G$ per grooming received, including events the individual did not choose.
- *No actor update.* P4 is a learned critic combined with softmax choice at the P2 temperature. With $\alpha_w = 0$ and $H_{\text{imm}} = H$ it reproduces P2 exactly.

P3D (deterrence-aware planning). This is P3-T plus a deterrence value on contest options and on responses to attacks.

- Individual i estimates j's forecast against it by reading its own record about j from j's side: $\tilde{p}_{ji} = (\alpha + l_{ij} + \omega_S s_{ij}^-)/(2\alpha + w_{ij} + l_{ij} + \omega_S(s_{ij}^+ + s_{ij}^-))$.
- It models j's challenge rate on its holdings as $\lambda_j = 2\tilde{p}_{ji}\hat{h}_i/(N-1)$.
- For each outcome o (j yields, i wins, i loses, i yields), $D(o) = -[\lambda'_j(o) - \lambda_j] H_d (1 - \hat{p}_{ij}) W_i$, with $H_d = (1 - e^{-(\kappa+\gamma)H})/(\kappa + \gamma)$.
- Q gains $\sum_o \hat{\pi}(o) D(o)$.

All of these inputs come from the individual's own records.

Reference policies. These are used only in comparison arms.

- **P0.** Uniform over feasible action types, then over targets and efforts.
- **P1.** Fixed hawk, dove or bourgeois rules.
- **C2.** No learning.

7.7 Exposure ledger

Physiological inputs come only from realized events and states. Every jump is a row of an auditable ledger with its time, individual, channel, magnitude and buffer factor. Continuous inputs are kept as exact cumulative totals. No rank appears in any row.

Channel	Trigger	Form	Magnitude	Pathway
defeat	lose a fight (either role)	jump	q_D	threat

Channel	Trigger	Form	Magnitude	Pathway
intimidation	yield: SUBMIT or failed ESCAPE, to CHALLENGE or AGGRESS	jump	q_S	threat
injury	injured in a fight	jump	q_I	threat
unpredictability	fight lost, injury or failed escape, scored against the individual's own forecast	jump	$\beta_U[-\ln \widehat{p}(\text{outcome})]$	threat
contest effort	take part in a fight, win or lose, with own effort e	jump	$q_E e$	effort
escape effort	escape attempt, success or failure	jump	q_{esc}	effort
activity	forage, withdrawn, holding, consorting, cooperating	continuous	$\beta_W(m - m_{\text{rest}})$	effort
guarding	consorting	continuous	$\beta_W m_G$	effort
deficit	$E_i = 0$ with deficit accruing	continuous	β_D	deprivation
anticipatory threat	holding an asset (off by default)	continuous	$\beta_T \max(\widehat{h}_i W_i, 0)/U_{\text{ref}}$	threat

Accounting rules.

- A fight gives one contest-effort jump to each participant and one defeat jump to the loser. The activity input is zero during fights and grooming, so no continuous term duplicates a fight's jumps.
- Each yield gives exactly one intimidation jump. A successful escape gives escape effort only.
- Effort inputs apply to winners as well (the cost-of-dominance pathway).
- *Unpredictability* is scored against the experiencing individual's own forecast at commitment:
  - a lost fight, $1 - \widehat{p}_{ij}(e_i, \widehat{e}_j)$;
  - an injury, $\iota\widehat{e}_j$;
  - a failed escape, $1 - \phi$.
- Fight and injury surprise apply only when the loser's direct evidence about the winner was at least n_U before the fight.
- Positive outcomes, chosen submissions, declined proposals, rejected matings and season-end losses are not scored.
- Threat-pathway inputs are multiplied by the support buffer $\beta_B(B_i)$ (§7.8). Effort and deficit inputs never are.
- The safety term s_S of §7.5 is a separate, decision-time aversion and is not a physiological input.

7.8 Physiology

Activation and load follow

$$\dot{z}_i = -\lambda_z z_i + u_i(t), \qquad \dot{A}_i = k z_i^p - \delta_A(B_i)(A_i - \bar{A}),$$

with jumps at events,

$$z_i(t^+) = z_i(t^-) + \beta_B(B_i)\, q^{\text{threat}} + q^{\text{effort}}, \qquad u_i = \beta_B(B_i) u_T + u_W + u_D.$$

The defaults are $p = 1$, $\lambda_z = 1 \gg \delta_{A0} = 0.01$, $\bar{A} = 0$, and $k = \lambda_z \delta_{A0}$. This choice of k makes the stationary mean excess load equal to the mean input rate. The support trace decays, $\dot{B}_i = -\lambda_B B_i$. It jumps by b_G when i receives grooming and by b_C for each cooperation partner, applied at acceptance.

Support. The two mechanisms are tested separately, and neither subtracts load directly.

- *Threat buffering:* $\beta_B(B) = 1 - \eta_B B/(K_B + B)$ scales threat inputs only.
- *Recovery facilitation:* $\delta_A(B) = \delta_{A0}[1 + \rho_B B/(K_B + B)]$.

Integration.

- *Exact engine* ($p = 1$, $\rho_B = 0$, piecewise-constant input). Between events the closed-form flow map is used:

$$z(\tau) = z_0 e^{-\lambda_z \tau} + \frac{u}{\lambda_z}(1 - e^{-\lambda_z \tau}),$$

$$A(\tau) = \bar{A} + (A_0 - \bar{A})e^{-\delta_A \tau} + k z_0 \frac{e^{-\delta_A \tau} - e^{-\lambda_z \tau}}{\lambda_z - \delta_A} + \frac{ku}{\lambda_z}\left[\frac{1 - e^{-\delta_A \tau}}{\delta_A} - \frac{e^{-\delta_A \tau} - e^{-\lambda_z \tau}}{\lambda_z - \delta_A}\right].$$

  These expressions and the window integrals of z and A are evaluated as divided differences of the exponential. Every coefficient is then positive, so $z, A \geq 0$ hold in floating point without clipping, and the equal-rate limit needs no separate branch. Intervals are split where E reaches 0, so the deficit input starts exactly.
- *Semi-analytic engine* ($p \neq 1$, $\rho_B > 0$, or buffering acting on a nonzero u_T). z is in closed form, and A is computed from its convolution form by adaptive quadrature at relative tolerance `physio_rtol`. The integral of $\delta_A(B(s))$ is in closed form.

Analytic properties. These were verified symbolically and, where stated in the main text, by machine-checked proof:

- with zero input, $z \to 0$ and $A \to \bar{A}$;
- under constant input, $z^* = u/\lambda_z$ and $A^* = \bar{A} + (k/\delta_A)(u/\lambda_z)^p$;
- a single pulse q gives $A(t) - \bar{A} = kq(e^{-\delta_A t} - e^{-\lambda_z t})/(\lambda_z - \delta_A)$;
- jumps at mean rate r give mean excess load $krq/(\lambda_z \delta_A)$, independent of their temporal clustering in the linear model.

An independent ODE solver with jump callbacks is used in tests only.

7.9 Optional mechanisms

Mating seasons and consortships (off by default, $K_M = 0$).

- Each period of length T_{season} starts with an in-season interval of length $s_M T_{\text{season}}$. During it there are K_M consortship assets and $r_i = 1$ for every individual. With $s_M = 1$ the breeding position is year-round.
- An idle consort makes mating attempts as a Poisson process of rate μ_m [mating], paused while it is busy. Each attempt is accepted with probability a (default 0.5), independently of contest history. Optionally, $a_i = \sigma(\text{logit}\, a + \chi(x_i - \frac{1}{2}))$, with x_i the objective decayed fight-win share.
- Consorting costs an extra m_G (guarding) and reduces intake to $\psi_M g_s$.
- At season end all consortships dissolve and their holders enter forage mode.

Fission–fusion (off by default, $n_{\text{sub}} = 1$).

- There are n_{sub} subgroups. Each individual moves at rate ω_{FF} to a uniformly chosen subgroup, possibly its own [fission].
- Encounters are restricted to co-present individuals. Visible assets are free assets and assets held by co-present individuals.
- Holdings, ongoing interactions and mating are unaffected by moves.

Membership turnover and targeted removal.

- *Random turnover.* At group rate ξ [shocks], a uniformly chosen individual is replaced by a naive newcomer. The newcomer gets the initial state, the prior beliefs and, under P4, zero critic weights. Every other individual's records about that slot are erased. Any asset it held becomes free, and an ongoing interaction ends for its partner.
- *Targeted removal.* This is a declared experimenter intervention. At `removal_time`, the individual with the highest David's score is replaced in the same way. The score is computed analyst-side from the dominance records of the preceding 500 time units, and ties are broken on the shocks stream. It is not an agent rule, and the score it uses enters no behavioural, belief or physiological equation.

Individual traits (off by default; Appendix S5). Each individual draws $z_1, z_2 \sim N(0, 1)$ on entry [traits]. - Fighting ability $g_i = \sigma_F z_1$ adds $g_i - g_j$ to the logit of P_{ij}. - Affiliative motivation $h_i = \sigma_S(r z_1 + \sqrt{1 - r^2}\, z_2)$ replaces s_G in the individual's own grooming values by $s_G e^{h_i - \sigma_S^2/2}$. - No individual observes any trait, its own fighting ability included. Neither trait enters physiology.

Rival-proximity vigilance (off by default; Appendix S6). An idle consort holder receives a threat jump q_V each time an eligible rival (in season, not consorting) encounters it, unless that rival's chosen action targets the holder.

Rank insecurity (off by default; Appendix S6). A continuous threat input $u_I = \beta_I \, \mathcal{U}_i$, where $\mathcal{U}_i$ is the mean of $4\hat{p}_{ij}(1-\hat{p}_{ij})$ over co-present partners with direct evidence $n_{ij} \geq n_U$. It is computed from the individual's own forecasts at each event touching it.

Two-way coupling (off by default; enabled one at a time). With all four mechanisms off, coupling is one-way: load cannot affect choices, effort or contest outcomes.

- **F1, private burden cost** (P4 only). At each TD step, $R \mathrel{-}= w_A \tilde{a}$. Here $\tilde{a} = (\int z\, dt) \exp(\sigma_a \epsilon - \sigma_a^2/2)$, the integral is taken over the inter-decision interval, and $\epsilon \sim N(0,1)$ is drawn on the observation stream. F1 has no channel into P2 or P3.
- **F2, load-limited effort capacity.** The feasible high effort is $e_L + (e_H - e_L)(1 - (A_i - \bar{A})/A_{\text{cap}})_+$, evaluated at decision or response time. The high option is dropped when it equals e_L. This is a capacity limit, not a sensed signal.
- **F3, load-slowed recovery.** $\delta_A(A,B) = \delta_A(B)/(1 + \chi_A(A - \bar{A}))$, integrated by adaptive Dormand–Prince 5(4), with z and B in closed form. This is the only mechanism under which temporal clustering of inputs can change mean load. The channel decomposition does not hold under F3.
- **F4, functional reserve.** The reserve follows

$$\dot{R}_i = \frac{\rho_R(1-R_i)}{1+\chi_R(A_i - \bar{A})} - \eta_R R_i \frac{z_i^m}{K_z^m + z_i^m}, \qquad R_i(0) = 1,$$

  and $R_i \in [0,1]$ is proved invariant. It caps the feasible high effort at $e_L + (e_H - e_L)R_i$. If F2 is also set, F4 takes precedence.

Table S1. Model parameters

Defaults are the values in the simulation code. "Values explored" lists settings used in the experiments: E-numbers are sweep configurations, A–D are the robustness base points and W/F/L are wildlife scenario cells. Base points: **A** $N = 10$, $\zeta = 2$, $m = 0.03$ ($m/g_s = 1.2$), $\gamma = 0.01$, $\tau_T = 0.1$, standing cue on, $\omega_{\text{obs}} = 0.3$. **B** $N = 6$, $\zeta = 2$, $m = 0.02$ (0.8), $\gamma = 0.001$, $\tau_T = 0.1$, cue off, $\omega_{\text{obs}} = 1$. **C** $N = 6$, $\zeta = 2$, $m = 0.03$ (1.2), $\gamma = 0.001$, $\tau_T = 0.03$, cue on, $\omega_{\text{obs}} = 1$. **D** base A at $N = 20$. All four bases have $s_S = 0.02$. "Morris" and "Sobol" give the ranges of the global sensitivity designs. "Fixed" means the parameter was not varied.

Symbol	Code name	Meaning	Default	Values explored
Group, contact and run				
N	`N`	group size	10	6, 10, 20 (E1, E1b; bases B, C at 6, D at 20); Morris 6–14
n_{enc}	`n_enc`	maximum encounters per opportunity	2	2, 4 (E1b_N20, C1 arm only)
n_{vis}	`n_vis`	visible assets per opportunity	2	fixed
n_{sub}	`n_sub`	number of subgroups (fission–fusion)	1 (off)	2, 4 (E12, W8a/b, F, L cells)
ω_{FF}	`omega_FF`	per-individual subgroup move rate	0	0.003, 0.03
T_{run}	`T_run`	run length	3000	9000 at $N = 20$; 6000 (E11 window convergence)
Δt_{rec}	`dt_rec`	recording-grid spacing	1	fixed
Energy and food				
g_s	`g_s`	dispersed intake rate	0.025	fixed
m	`m`	maintenance rate	0.02 ($m/g_s =$ 0.8)	$m/g_s = 0.5, 0.7, 0.9, 1.0, 1.1, 1.2$ (E3); 0.8, 1.0, 1.2 (E1b); 1.2 (bases A, C); 0.9 (W3a, W4); Morris and Sobol 0.5–1.2

Symbol	Code name	Meaning	Default	Values explored
m_{rest}	`m_rest`	expenditure while resting	0.01	fixed
E_0	`E0`	initial energy	0.5	fixed
E_h	`E_h`	curvature of the food value Φ	0.3	fixed
K_F	`K_F`	number of food patches	3	$K_F/N = 0.1$, 0.2, 0.3, 0.5, 0.7, 1.0 (E3); 0.1, 0.3, 1.0 (R1); 1 (W5b, W9a); 0 (E5 mating only); Morris and Sobol K_F/N 0.1–1.0
ρ_c	`rho_c`	patch richness, g_P/g_s	4	1, 2, 4, 6, 8 (E3); 1, 2, 4, 8 (R1); 1.5–8 (E2); 10 (W5b, W9a); set by matched supply $G_{\text{tot}} = 0.475$ ($N = 10$), 0.285 ($N = 6$), 0.95 ($N = 20$); Morris and Sobol 1–8
$\hat{\rho}_c$	`perceived_rho_c`	patch richness as perceived by agents	$= \rho_c$	2, 6 (E8)
Contest				
θ	`theta`	contest decisiveness	1	Morris 0.5–3
ζ	`zeta`	condition dependence of contest success	1	0, 0.5, 1, 2 (E1); 0, 2 (E1b); 2 (all bases); Morris 0–3
ε_c	`eps_c`	condition offset	0.1	fixed
e_L, e_H	`e_L`, `e_H`	low and high effort	0.25, 1	fixed
c_e	`c_e`	energy cost per unit effort	0.02	0, 0.01, 0.02, 0.05, 0.1 (E3); 0, 0.02, 0.1 (R1, E2); Morris and Sobol 0–0.1
ι	`iota`	injury probability per unit winner effort	0.2	fixed
E_{inj}	`E_inj`	energy lost per injury	0.05	fixed
d_F	`d_F`	fight duration	0.2	fixed
β_s, Δ	`beta_s`, `Delta`	C1 status slope and update step	1, 0.5	fixed (C1 arms only)
Escape, refuge and avoidance				
ϕ	`phi`	escape success probability	0.5	0, 0.2, …, 1.0 (E3); 0, 0.4, 0.8, 1.0 (R1); 0, 0.5, 0.9 (E2); 0 (W1, W2, W5b, W9a); Morris and Sobol 0–1
c_{esc}	`c_esc`	energy cost of an escape attempt	0.005	fixed
ϕ_W	`phi_W`	refuge protection from encounters	0.9	0, 0.3, 0.6, 0.9 (E3); 0 (E4); 0.1 (W1, W2); Morris 0–1
ψ_W	`psi_W`	relative intake while withdrawn	0.5	fixed
τ_V	`tau_V`	avoidance duration after a defeat or yield	20	fixed
v	`v`	reduction in contact weight under avoidance	0.8	fixed
Affiliation, cooperation and utility				
d_G	`d_G`	grooming duration	0.5	fixed
s_G	`s_G`	utility of received grooming	0	0, 0.02 (E6)
d_C	`d_C`	cooperation duration	1	fixed

Symbol	Code name	Meaning	Default	Values explored
σ_c	`sigma_c`	cooperative intake gain	0.2	1.0 (E6, W9b); Morris 0–1
s_H	`s_H`	subjective harm of an injury	0.05	fixed
s_S	`s_S`	subjective cost of a received safety event	0.02	0, 0.02 (E1b, E1c); Morris and Sobol 0–0.1
Beliefs				
α	`alpha`	Beta prior strength (each side)	1	fixed
γ	`gamma`	decay rate of evidence	0.01	0.001, 0.01, 0.1 (E1); 0.001, 0.01 (E1b); 0.001 (bases B, C); Morris and Sobol 0.001–0.1 (log)
ω_S	`omega_S`	weight of yields relative to fights in $\hat{p}$	0.5	fixed
ω_{obs}	`omega_obs`	bystander pseudo-count weight	0 (off)	0, 0.3 (E1); 0, 0.3, 1.0 (E1b); 0.3 (bases A, D), 1.0 (B, C); Morris 0–1
—	`standing_cue`	own-experience standing cue	off	off, on (E1b); on (A, C, D), off (B); Morris binary
β_{opt}	`beta_opt`	optimism bias in win forecasts	0	0, 1 (E8)
α_R	`alpha_R`	Dirichlet prior strength of response forecasts	3	fixed
n_U	`n_U`	evidence below which a dyad is unobserved	1	fixed
a_h, b_h	`a_h`, `b_h`	Gamma prior of attack rates (shape, rate)	1, 10	a_h = 1, 2, 4 (E1c diagnostics)
—	`learning`	belief learning (false = C2)	true	false (C2 arms)
—	`dyadic_learning`	partner-specific learning (false = P2-pooled)	true	false (E1c)
Policy				
—	`policy`	P0, P1, P2 or P4 (P3, P3-T and P3D are P2 settings)	P2	P0, P1 (E1); P3, P4 (E9); P3-T, P4b (E9b); P3D (E13); P3-T (W5b, W5c)
τ_T	`tau_T`	relative choice temperature	0.1	0.03, 0.1, 0.3 (E1); 0.03, 0.1 (E1b); 0.03 (base C); Morris and Sobol 0.01–1 (log)
H	`H`	planning horizon	10	100 (P3, P3-T, P3D; E8)
T_H	`T_H`	horizon in the temperature scale	$= H$	10 (P3-T, P3D)
κ	`kappa`	discount rate	0.01	fixed
H_{imm}	`H_imm`	P4 immediate horizon	1	fixed
α_w	`alpha_w`	P4 normalized critic step size	0.05	fixed
—	`critic_beliefs`	P4b mean-win-forecast feature	off	on (P4b)
—	`deterrence`	P3D deterrence value	off	on (P3D)
Exposure ledger				
q_D	`q_D`	defeat jump	1	0 (ablation: E4, R1); Morris 0–2
q_S	`q_S`	intimidation jump	0.5	0 (ablation: E4, R1)
q_I	`q_I`	injury jump	2	0 (ablation: E4)
q_E	`q_E`	contest-effort jump per unit effort	0.5	0 (ablation: E4, R1)
q_{esc}	`q_esc`	escape-effort jump	0.2	0 (ablation: E4)
β_W	`beta_W`	activity-effort gain	10	0 (ablation: E4)
β_D	`beta_D`	deficit input rate	0.5	0 (ablation: E4, R1); Morris 0–1
β_U	`beta_U`	unpredictability gain (per nat)	0.5	0 (ablation: E4)

Symbol	Code name	Meaning	Default	Values explored
β_T	`beta_T`	anticipatory-threat gain	0 (off)	0, 0.5 (E8)
Physiology and support				
λ_z	`lambda_z`	activation decay rate	1	fixed
δ_{A0}	`delta_A0`	baseline load recovery rate	0.01	fixed
k	`k`	load gain	$\lambda_z\delta_{A0} = 0.01$	fixed
p	`p`	activation exponent in the load equation	1	fixed
$\bar{A}$	`Abar`	baseline load	0	fixed
λ_B	`lambda_B`	support-trace decay rate	0.05	fixed
b_G, b_C	`b_G`, `b_C`	support jumps from grooming and cooperation	1, 0.5	fixed
K_B	`K_B`	support half-saturation	1	fixed
η_B	`eta_B`	threat buffering by support	0 (off)	0, 0.8 (E6, R1, W2); 0.6 (E11 solver convergence); Morris 0–1
ρ_B	`rho_B`	recovery facilitation by support	0 (off)	0, 0.8 (E6, R1, W2); Morris 0–1
—	`physio_rtol`	relative tolerance of the adaptive engines	10^{-10}	10^{-6}, 10^{-8}, 10^{-10}, 10^{-12} (E11 solver convergence)
Two-way coupling (Stage 5)				
w_A	`w_A`	F1 burden cost (P4 reward only)	0 (off)	0.005, 0.02 (E10)
σ_a	`sigma_a`	F1 log-normal sensing noise	0.5	fixed
A_{cap}	`A_cap`	F2 load scale of effort capacity	∞ (off)	1.5, 0.75 (E10, R1)
χ_A	`chi_A`	F3 load slowing of recovery	0 (off)	1, 5 (E10, R1)
—	`reserve`	F4 functional reserve on	off	on (E14)
ρ_R	`rho_R`	F4 reserve recovery rate	0.05	0.02, 0.1 (E14)
χ_R	`chi_R`	F4 load slowing of reserve recovery	1	fixed
η_R	`eta_R`	F4 reserve depletion rate	0.2	0.1, 0.5 (E14)
K_z, m	`K_z`, `m_R`	F4 activation half-saturation and Hill exponent	1, 2	fixed
Mating				
K_M	`K_M`	consortship assets per season	0 (off)	1, 2, 5 (E5); 1 (E9, E13; W4, W5a, W5c, W7)
T_{season}	`T_season`	season period	200	1000 (W7)
s_M	`s_M`	in-season fraction of the period	0.25	0.25, 1.0 (E5); 1.0 (W4, W5a, W5c); 0.5 (W7)
μ_m	`mu_m`	mating-attempt rate of a consort	0.5	fixed
a	`a_mate`	mate-acceptance probability	0.5	fixed
ψ_M	`psi_M`	relative intake while consorting	0.5	fixed

Symbol	Code name	Meaning	Default	Values explored
m_G	`m_G`	guarding cost while consorting	0.01	0, 0.01, 0.03 (E5); 0.03 (W4, W5a, W5c, W7)
s_M^u	`sM_u`	utility of an accepted mating	0.1	fixed
χ	`chi`	winner preference in mate acceptance	0	0, 2 (E5)
Membership				
ξ	`xi`	group turnover rate	0	0.001, 0.003, 0.01 (E7); 0.01 (W6b); Morris 0–0.01
—	`removal_time`	time of the targeted removal	∞ (off)	2000 (E7); 1500 (W6a)
—	`removal_window`	dominance records used to rank before removal	500	fixed
Fitness proxies (recorded only)				
μ_0, μ_D, μ_I	`mu_0`, `mu_D`, `mu_I`	baseline, deficit and per-injury hazards	10^{-4}, 10^{-2}, 0.05	fixed

Appendix S2. Verification

Verification ran on three levels, and every level is part of the archived repository.

Table S2. Symbolic derivations (SymPy; `symbolic/tests/`). The Julia decision formulas and reference values are generated from these derivations and checked against them.

Item	Content
S1	Flow map by variation of constants: ODE residual zero, initial condition, equal-rate limit, semigroup property
S2	Exact window integrals of z and A
S3	Steady state, Jacobian eigenvalues, single-pulse response; Campbell mean and variance for Poisson and compound-Poisson input
S4	Periodic-average identity with jumps at arbitrary times
S5	Integrating factor for recovery facilitation with decaying support
S6	Energy ledger: hitting times of 0 and 1; piecewise deficit and waste
S7	Asset holding value and all decision values, including the limits $\widehat{h} \to 0$, $H \to \infty$, $\kappa \to 0$
S8	Contest success function: complementarity, monotonicity, homogeneity; Tullock first- and second-order conditions
S9	Softmax: normalization, shift invariance, low- and high-temperature limits
S10	Beta and Dirichlet forecasts with forgetting: bounds; prior mean 1/2 for an empty dyad
S11	Mean-field load decomposition by channel
S12	Exact enumeration of expected Landau h and the cyclic-triad distribution for small N
S13	Individual traits: lognormal mean correction; the logit-additive ability term equals Tullock strength $c_i e^{g_i}$; trait covariance $r\sigma_F\sigma_S$

Item	Content
S14	Insecurity term $4p(1-p) \leq 1$ with maximum at $\frac{1}{2}$; steady excess load of a constant insecurity input

Table S3. Machine-checked proofs (Lean 4 with Mathlib (The mathlib Community 2020); `lean/AllostaticHierarchy/`). The build contains no unproved statement (`sorry`).

Item	Theorem(s)	Statement
L1	`flow_nonneg`, `flow_A_nonneg`	the closed-form $z(\tau)$ and $A(\tau)$ are nonnegative under nonnegative input, without clipping
L2	`flow_hasDerivAt`, `flow_initial`	the closed forms satisfy the ODEs and the initial condition
L3	`flow_semigroup`, `steady_state_unique_fixed_point`	the flow map is a semigroup with a unique fixed point
L4	`period_average`	for periodic solutions, mean load equals $\bar{A}$ plus k/δ_A times mean activation
L5	`jensen_clustering`	finite Jensen inequality for x^p, $p \geq 1$ (clustering can raise mean load only if $p > 1$)
L6	`softmax_simplex`	softmax probabilities are positive and sum to 1
L7	`csf_props`, `csf_strictMono_effort`	$P_{ij} \in (0,1)$, $P_{ij} + P_{ji} = 1$, strictly increasing in own effort
L8	`ledger_conservation`, `ledger_bounds`	an executable specification of the per-event energy update conserves the ledger identity over any finite event list and keeps $E \in [0,1]$
L9	`beta_forecast_bounds`, `beta_forecast_empty`	win forecasts lie in $(0,1)$; empty evidence gives $1/2$
L10	`scalar_order_acyclic`, `scalar_no_three_cycle`	the scalar comparator C1 induces a strict weak order, so it cannot produce expected cycles
L11	`tournament_transitive_iff_no_3cycle`	a tournament is transitive if and only if it has no directed 3-cycle
L12	`landau_nonneg`, `landau_le_one`, `landau_eq_one_iff`	Landau's $h \in [0,1]$, with $h = 1$ exactly for the transitive tournament
L13	`physioField_lipschitz`, `physio_solution_unique`, `solution_eq_flow`	the physiology vector field is Lipschitz, so solutions are unique, and every solution equals the closed-form flow map
L14	`reserve_le_one`, `reserve_nonneg`, `f4_reserve_invariant`	the functional reserve of F4 stays in $[0,1]$
L15	`half_le_csf_iff`, `csf_strong_transitive`	contests decided by fixed abilities are strongly stochastically transitive, so an order produced by ability alone is transitive by construction
L16	`insecurity_term_bounds`, `insecurity_input_bounds`	the insecurity input lies in $[0, \beta_I]$, so the invariant domain of L1 is preserved

Automated tests (Julia). 15,090 tests in thirteen groups: random-stream independence and reproducibility; package quality (Aqua); agreement with the generated symbolic references; physiology (analytic tests and

comparison with an independent ODE solver with jump callbacks); the game (ledger identity after every event, feasibility, invariants, controls); exposure (exact decomposition of load into channels, and a test that load never reaches decisions under one-way coupling); structure analysis; rank–load and profile-shape rules; two-way coupling; sensitivity estimators; the stage-game benchmark; individual traits; and the vigilance and insecurity inputs.

For the trait, vigilance and insecurity extensions, the full set of earlier wildlife scenarios was re-run with the extensions switched off. Every recorded output was bit-identical to the original runs.

Appendix S3. Supplementary results

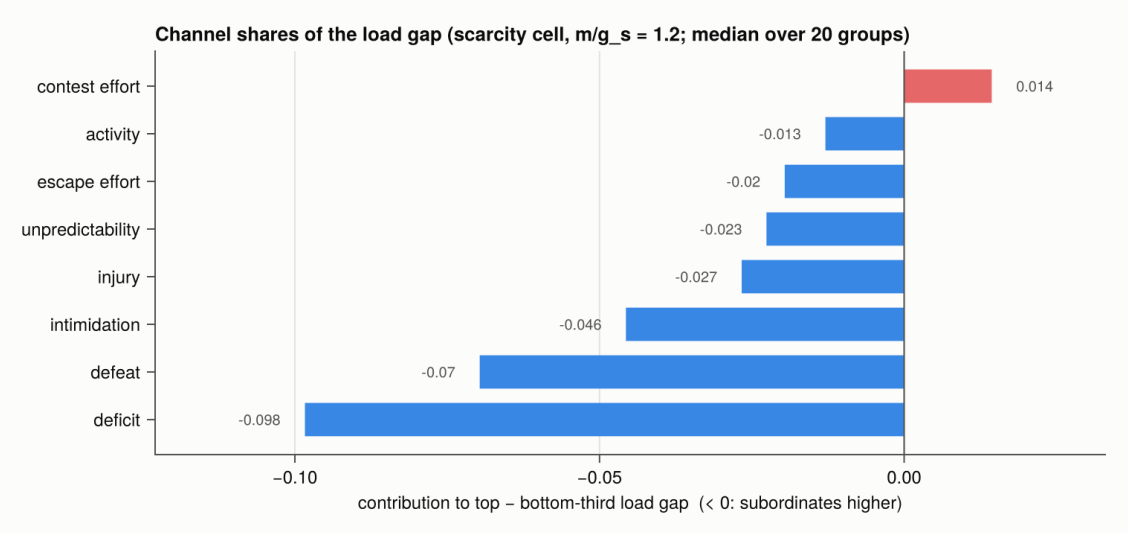


Figure 1: **Figure S1.** Channel attribution of the top-versus-bottom-third load gap at the scarcity cell of base A ($m/g_s = 1.2$; median over 20 groups). Negative contributions mean subordinates carry more load through that channel. Energy deficit and defeat dominate; contest effort is the only channel loading dominants more.

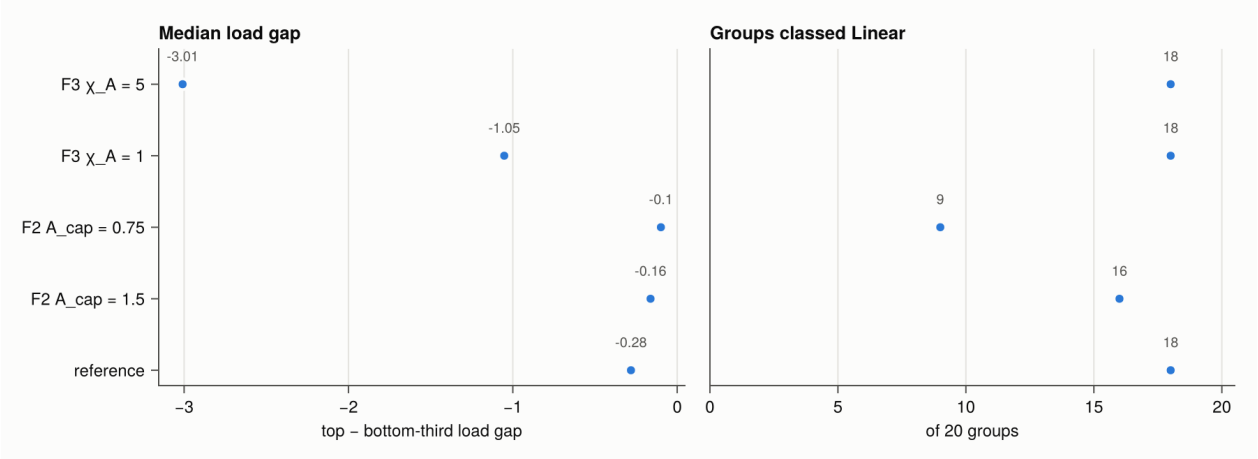


Figure 2: **Figure S2.** Two-way coupling, one mechanism at a time, at base A. Left: median top-minus-bottom-third load gap. Right: number of 20 groups classed Linear. Load-slowed recovery (F3, χ_A) amplifies the gap without changing structure; load-limited effort capacity (F2, A_{cap}) shrinks it, and at $A_{\text{cap}} = 0.75$ halves the number of Linear groups.

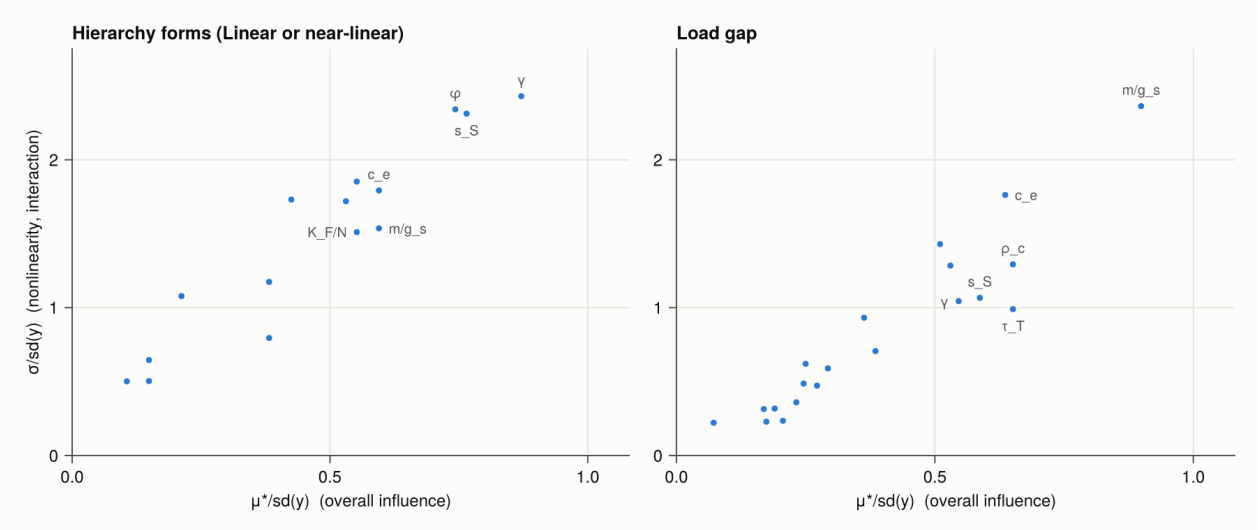


Figure 3: **Figure S3.** Morris screening over the ranges in Table S1, for whether a hierarchy forms (Linear or near-linear; left) and for the load gap (right). Both axes are scaled by the output's standard deviation. Points far above the diagonal act mainly through nonlinearity or interaction.

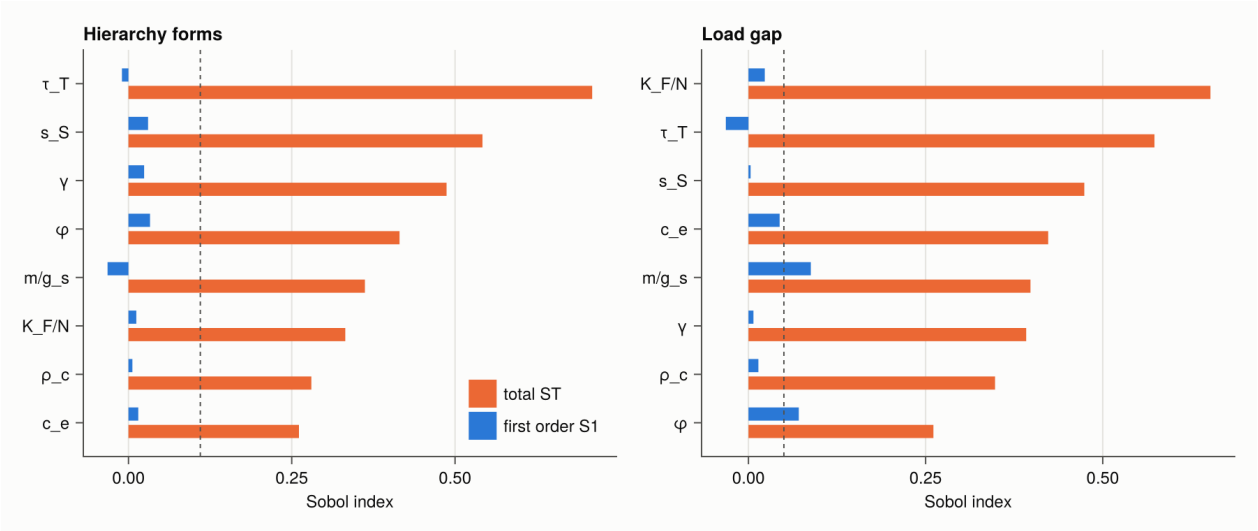


Figure 4: **Figure S4.** Sobol first-order (S_1) and total (S_T) indices for the eight parameters selected by Morris screening. Dashed line: the replicate-noise floor. First-order indices are near zero and total indices large, so outcomes are governed by interactions among parameters.

Appendix S4. Group size and the contender contrast

Design. The shape results in the main text come from groups of 6 and 10. Closed groups of 20 do not order at base A, so the $N = 20$ tests used the settings known to order at that size: fast fission–fusion (four subgroups, move rate 0.03) for the base-A scenarios, with matched $N = 10$ fission–fusion controls, and bases B and C. Predictions were fixed before the runs. Windows were three times longer at $N = 20$ (burn-in 3,000, windows of 1,500), and each cell had 40 groups.

Table S4. Shapes of the load profile at $N = 20$ and in matched $N = 10$ controls (FF: fission–fusion).

Cell	Setting	N	Ordered /40	Shape (thirds rule)	Prediction
L0	base A, closed	20	0	–	fewer than 5 ordered: holds

Cell	Setting	N	Ordered /40	Shape (thirds rule)	Prediction
L1 / F0	base A + FF	20 / 10	40 / 35	subordinate-high, graded / same	subordinate-high: holds
L2 / F1	as L1, no escape	20 / 10	39 / 38	subordinate-high / same	subordinate-high: holds
L3	base B	20	0	–	untestable
L4	base C	20	6	subordinate-high, alpha trough	holds
L5 / F4	ample food, guarded breeding monopoly + FF	20 / 10	2 / 15	– / subordinate-high, alpha trough	untestable
L6 / F5	mating monopoly + FF	20 / 10	9 / 31	subordinate-high, alpha trough / same	subordinate-high, no alpha peak: holds
L7 / F9	tolerant contests + FF	20 / 10	0 / 9	– / indeterminate	no prediction
K14	base B, $\tau_T = 0.03$	20	31	flat, alpha trough	see Table S5
K18	base C, $m/g_s = 0.8$	20	39	dominant-high, alpha trough	see Table S5

Re-running the earlier $N = 20$ designs to add profile records reproduced every previously recorded output bit for bit. In those designs, every setting without scarcity that ordered at least 12 of 20 groups had its mean profile peak at rank 2 or 3, above a flat remainder and below a low-load alpha. The thirds rule read these profiles as dominant-high or, once, as flat, because at $N = 20$ ranks 2–3 lie inside the top third with the alpha.

Contender contrast. To describe this profile at any group size we defined, after the scan and before a confirmation run on new seeds (K14, K18), the contender contrast

$$\kappa = \tfrac{1}{2}\left(\ell_{(2)} + \ell_{(3)}\right) - \text{mean}\left(\ell_{(5)}, \ldots, \ell_{(N)}\right),$$

on the same standardized, score-ordered profile as the shape rule. Position 4 is excluded as a buffer. The confirmation prediction was ≥ 5 ordered groups, a κ interval above 0.5 and an alpha trough, in both cells. It held.

Table S5. Contender contrast κ and alpha contrast α_c (95% bootstrap intervals over groups). The values for E15 cells were computed after those runs and are post hoc.

Cell	N	Ordered	κ	α_c	Shape (thirds rule)
K14 (confirmation)	20	31/40	1.23 (0.84, 1.62)	−1.53 (−2.25, −0.78)	flat, alpha trough
K18 (confirmation)	20	39/40	1.45 (1.06, 1.79)	−2.13 (−2.68, −1.54)	dominant-high, alpha trough
W3b (post hoc)	6	25/40	1.53 (1.36, 1.69)	−1.63 (−1.85, −1.38)	inverted U, alpha trough
W3a (post hoc)	10	7/40	1.30 (0.95, 1.63)	−0.92 (−1.65, −0.22)	indeterminate
W9b (post hoc)	10	9/40	0.89 (0.37, 1.36)	−0.13 (−0.75, 0.50)	dominant-high
W4 (post hoc)	10	37/40	−0.29 (−0.50, −0.09)	−1.37 (−1.62, −1.13)	inverted U (middle peak)
W9c (post hoc)	6	27/40	0.02 (−0.09, 0.14)	−1.99 (−2.13, −1.84)	subordinate-high, step

Cell	N	Ordered	κ	α_c	Shape (thirds rule)
W0 (post hoc)	10	38/40	−1.42 (−1.51, −1.32)	−0.23 (−0.33, −0.12)	subordinate-high, graded

At the 31 base-B scan points with at least 5 ordered groups (post hoc), κ exceeded 0.9 at all 19 points with three patches of richness $\rho_c \geq 4$. These include the channel ablations, support and coupling variants of that cell. κ was at most 0.57 at the 12 points with poor patches ($\rho_c \leq 2$) or in the matched-supply design of patch number and escape. The contender peak therefore requires several rich, contestable patches without scarcity.

Appendix S5. Individual differences in fighting ability and affiliation

Design (declared before the runs; 40 groups per point; common random numbers, so a replicate has the same trait ranks at every spread). - **Part A:** fighting-ability spread $\sigma_F \in \{0, 0.25, 0.5, 1, 2\}$ in five cells: - GW0 (scenario W0); - GZ (W0 with $\zeta = 0$); - GW4 (W4); - GW3b (W3b, $N = 6$); - GK18 (K18, $N = 20$). - **Part B:** affiliative-motivation spread $\sigma_S \in \{0, 0.5, 1\}$, alone and with $\sigma_F = 1$ at $r \in \{-0.5, 0, 0.5\}$. Cells are S2 (W2 with $s_G = 0.02$) and S2ns (S2 with support off). - **Statistics:** ρ_F and ρ_S are Spearman correlations of the trait with David's score in each group's first ordered window. β coefficients come from a within-group regression of standardized load on standardized rank position and the traits.

Table S6. Fighting ability (Part A). Means with 95% bootstrap intervals over ordered groups.

Cell	σ_F	Ordered /40	ρ_F	β_F	κ	Shape (thirds rule)
GZ	0 / 0.5 / 2	0 / 16 / 40	– / 0.87 / 0.96 (0.95, 0.97)	– / −0.17 / −0.61 (−0.73, −0.50)	–	– / subordinate-high / subordinate-high, alpha trough
GW0	0 / 0.25 / 1 / 2	40 / 40 / 40 / 40	– / 0.69 / 0.91 / 0.93	– / −0.05 / −0.25 / −0.41 (−0.53, −0.27)	−1.49 / −1.69 / −1.71 / −1.75	subordinate-high at every level
GW4	0 / 2	40 / 40	– / 0.95	– / −0.57 (−0.87, −0.22)	−0.19 / −1.06	inverted U, alpha trough (both)
GW3b	0 / 2	28 / 39	– / 0.93	– / −0.58 (−1.01, −0.08)	1.37 / 1.25	inverted U, alpha trough (both)
GK18	0 / 1 / 2	40 / 37 / 31	– / 0.68 / 0.77	– / −0.22 / −0.28 (−0.36, −0.20)	1.18 / 1.06 / 0.71 (0.40, 1.02)	dominant-high → flat, alpha trough

Table S7. Affiliative motivation (Part B; $\sigma_S = 1$ unless stated).

Point	ρ(motivation, grooming received)	ρ_S	β_S
S2, $\sigma_F = 0$	0.54 (0.46, 0.63)	−0.37 (−0.45, −0.29)	−0.12 (−0.21, −0.02)
S2ns (support off), $\sigma_F = 0$	0.54	−0.37	−0.12 (−0.22, −0.03)
S2, $\sigma_F = 1$, $r = -0.5$ / 0 / 0.5	0.60 / 0.67 / 0.75	−0.53 / −0.23 / 0.22	−0.15 / −0.12 / −0.17

Of 11 predictions, 9 held. Two failed: - unpredictability rose rather than fell with ability spread in GZ; - κ did not fall in GW3b.

The full tables are in `reports/E16g_results.md` of the archived repository.

Appendix S6. Candidate mechanisms for the patterns not reproduced

Design (mechanisms chosen and predictions declared before the runs; 40 groups per point). - **T0** (no new mechanism): mating utility $\in \{0.1, 0.5, 2\}$ × winner preference $\chi \in \{0, 1\}$, in W5a and W7. - **T2:** vigilance $q_V \in \{0, 0.07, 0.14, 0.28\}$ in W5a, W7 and W5c. The grid was scaled by a declared pilot of the rate at which consorts meet rivals (1.4 per unit time). - **I1:** insecurity gain $\beta_I \in \{0, 0.1, 0.3\}$ in W0, W6a, W6b and W3b. - Both new inputs change physiology only, so behaviour is identical across their gain levels. - **J-shape rule:** an α_c interval above 0.5, and the alpha's standardized load above the bottom third's mean.

Results. - **T0:** the J-shape rule was met at none of the 12 points. - **T2:** - In W5a and W7 the top-ranked individual held the consortship only 3–12% of consort time, and vigilance deepened the alpha trough. - In W5c the top held it 48% of the time and α_c rose from -2.31 to -2.05 over the declared grid. - An exploratory run outside the grid (same seeds) met the J-shape rule at $q_V = 3$ ($\alpha_c = 1.92$, 1.52–2.29) but not at $q_V = 1$. - **I1:** - The insecurity input was nearly uniform across ranks: 0.18 for the alpha against 0.21 for the others at $\beta_I = 0.3$. - It did not strengthen the contender peak. - After the alpha's removal, mean load was lower in window 2 than in window 1 in 24 of 40 groups at every β_I, so the instability rule was not met.

The full tables are in `reports/E16_results.md`.

Appendix S7. Post hoc tests of explanations

Each explanation offered for a model pattern was tested, together with its corollaries, in exploratory runs after the results. Explanations that could not be tested were removed. The tests are listed in `reports/explanation_checks.md`. Those used in the main text:

- **H8, the low-load alpha.** In W5a and W9c (20 groups each), compared with ranks 2–3, the top individual:
  - fought less;
  - received less defeat and intimidation input (W5a: 0.04 against 0.11 per unit time);
  - spent less contest effort;
  - received far more yields than it gave (0.34 against 0.04).
- **H9, the contender peak.** In W3b, ranks 2–3 fought most (0.112 per unit time, against 0.100 for the alpha and 0.042 for the rest), and forecasts within the top four were least settled ($|\hat{p} - \frac{1}{2}| = 0.20$ against 0.28). Less-settled forecasts among the top four also occurred in W0, which has no contender peak, so this corollary is not specific to it.
- **H1, who holds the mate.** In W5a the top individual held food patches 61% of the time (others 16%). Making patches poorer raised its share of consort time from 0.07 to 0.17 and 0.33. A higher mating value did not.
  - A patch holder cannot occupy or challenge another asset. The top individual was free at only 1–3% of its decisions, left its patch at 0.3–0.5% of them, and took a visible consortship 67–83% of the time when free.
  - Allowing holders to challenge for the consortship directly (a counterfactual only) raised the top's share from 0.04–0.08 to 0.22–0.24.
  - Above a mating value of 0.5, starving individuals challenged the consortship continually, even at about 10% win odds. Mean tenure fell from 10 to 2.5 time units, and 97% of tenures ended in a lost fight. This is why a higher mating value did not raise the top's share.
- **H5, support.** Support scaled every individual's load nearly in proportion: profile correlation 0.996 with and without support.
- **H6, sociability and rank.** Controlling for energy halves the association between motivation and low rank. Counterfactual runs (identical except for one decision term) show that energy is not a separate route:
  - with motivation removed from the proposer's grooming value only, the association vanished ($\rho = -0.04$);
  - with motivation removed from the recipient's acceptance only, it stayed (-0.36);
  - making grooming cost no time weakened it but did not remove it (-0.26).

  Motivated individuals replaced unprovoked aggression with grooming proposals (aggression share 0.19 against 0.04 across motivation thirds; verified independently). Aggression won by the aggressor makes

up 43% of dominance records. An earlier statement that motivated individuals "lose fewer fights at the same rank" came from conditioning on David's score. Without that conditioning they lose a larger share of their fights (+0.23).

- **HA, patch exclusion.** In W0, patch-holding time fell from 0.46 in the top third to 0.04 in the bottom third, and deficit input rose from 0 to 0.095 per unit time. Within groups, patch time and deficit were negatively related (median $\rho = -0.86$). With seven patches, holding was more widely spread and the deficit gradient nearly vanished.
- **HB, two-way coupling.** Under load-limited capacity and reserve depletion, against the same seeds with the mechanism off, the most loaded third reduced high-effort fighting most (-0.20 and -0.17, against -0.09 and -0.06 for the least loaded). Yielding rose at every load level, slightly more among the least loaded. So "loaded individuals yield rather than escalate" holds for escalation but not for yielding.
- **H7, forecasts.** In W3b, median forecasts matched fight outcomes (difference -0.01), and dominance records including yields were more one-sided (0.84 against 0.67).
  - Calibration slopes show this agreement is a cancellation: W3b forecasts are overconfident (slope 0.77; 0.96 without yield evidence), while base-A forecasts are mildly underconfident (1.10).
  - At base A the Beta prior (weight 2) outweighs the decayed fight evidence (median 1.7). Weakening the prior (= 0.2) or slowing forgetting (= 0.003) closes the gap to fight outcomes.
  - The decayed counts were reconstructed from the logs and match the model's to 10^{-13}, so this is not a bookkeeping error.
- **Unpredictability with ability differences (GZ).** Per lost fight, surprise fell as ability spread grew. But the low-ability half kept attacking, and took more injuries and failed escapes, whose surprise does not depend on forecasts. Without injuries ($\iota = 0$) the rise persisted, carried by failed escapes.

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